

# Exact Truncation, Nilpotent Reduction, and Fast Terminal Tests

for Unknot Recognition

A proved and reproducible continuation toward  
quasi-polynomial algorithms

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**What this package establishes.** An exact scanner for selected homological degrees, a finite algebraic construction of minimal complexes, and a deterministic near-linear terminal boundary-pattern algorithm. The implementation is tested against the pinned ProveIt code and independent certificate identities.

**What remains open in the implementation.** The complete recognizer retains an exponential worst-case upper bound. The quasi-polynomial theorem proved here concerns prescribed homological windows under explicit diagram parameters; a general recognition bound requires an additional geometric or homological witness theorem.

Audited ProveIt revision:

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## Abstract

We continue the ProveIt unknot-recognition program with three concrete algorithms and an explicit analysis of the remaining complexity gap. First, we compute an exact prescribed interval of Khovanov homology by discarding partial chain summands before allocation whenever the remaining crossings cannot carry them into the target or its adjacent differential degrees. For a diagram with  $n$  crossings, frontier size  $w$ ,  $s$  Seifert circles, and minority crossing-sign count  $\mu$ , the delivered algorithm computes normalized degrees  $[-d, d]$  in  $2^{O(w+s+\mu+d)}(n+1)^{O(\mu+d+1)}$  bit operations. This gives an  $n^{O(\log n)}$  bound for those homological queries when  $w+s = O(\log^2 n)$  and  $\mu+d = O(\log n)$ . A mismatching window is a sound knot certificate; an agreeing window invokes the complete fallback. Second, the positive-degree ideal of the matching category has nilpotence exponent at most  $w+1$ , independently of object multiplicity. Binary matrix reduction followed by a finite perturbation series constructs an exact minimal complex and explicit deformation-retraction maps. Third, a known ball's trivalent boundary pattern is tested in  $O(s_P \log s_P)$  deterministic RAM operations by detecting dual loops, bigons, and nonfacial triangles; negative results carry independently verifiable primal-cut witnesses. On the supplied 36-crossing stress input, the window reduces the peak allocation from 17,694 to 7,692 objects and decides nontriviality under a 10,000-object ceiling that stops the full scanner. The paper includes paired measurements, proofs, integration code, an output-size obstruction for expanded normal-surface cuts, and a prioritized research program.

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# 1 Scope, source audit, and principal results

The task is to improve the existing recognizer while pursuing a quasi-polynomial general algorithm. Both objectives matter. A useful optimization must preserve the exact mathematical meaning of a verdict, and a complexity claim must count every object, coefficient, intermediate state, and output required by the implemented procedure.

The audited source is `VladimirReshetnikov/ProveIt`, revision `ca61a1a5f967999d778d6efd7c882e77a2eb379d`, specifically `Topology/UnknotRecognition` [1]. The active `fast/` code combines diagram validation, elementary reductions, connected-sum factorization, Alexander and Jones obstructions, and an exact  $\mathbb{F}_2$  Khovanov scanner. The same revision also stores the recent structural-compression continuation in `reports/07`; its patch has not been applied to the active `fast/` tree. This package’s integration patch targets that active tree. The baseline file hashes, source revision, new modules, and test output are supplied.

Recent work already supplied structural Seifert-graph tests, component sharing with exact multiplicities, and faster auxiliary filters. Other recent continuations developed a three-braid backend and exposed the distinction between frontier size and intrinsic complex multiplicity. The present work addresses different operations: which degrees are constructed, how a binary contraction is lifted to the full category, and how terminal small-cut searches are replaced by planar triangle enumeration.

| Result                   | Precise guarantee                                                                | Delivered status                                                            |
|--------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Moving degree window     | Exact requested homology after arbitrary valid scan order and unit cancellations | Implemented; complete fallback preserved                                    |
| Parameterized cost       | Uniform bit bound in $w, s, \mu, d, n$                                           | Proved for the implemented representation                                   |
| Finite nilpotent lift    | Exact minimal complex and checked chain maps                                     | Research backend; measured separately                                       |
| Terminal pattern test    | $O(s_P \log s_P)$ RAM time on a known ball                                       | Implemented; independent negative verifier                                  |
| Expanded-cut obstruction | $\Omega(W)$ output for $W$ explicitly listed parallel cut components             | Proved; informs the required interface                                      |
| General recognition      | Complete Khovanov fallback                                                       | Exponential upper bound; general quasi-polynomial implementation unresolved |

Table 1: Claims are separated by the mathematical quantity actually computed.

## 1.1 The literature establishes the boundary of the claim

Bar-Natan’s local cancellation framework is the foundation of the existing scanner [2]. Kelomäki and Schütz prove that explicit scanning and divide-and-conquer computations can be exponential even for positive or alternating three-braids. Their paper also gives a polynomial algorithm for full three-braid homology and polynomial algorithms for extremal degrees at fixed strand number [3]. These statements are compatible: an efficiently computed compressed answer can have an exponentially large basis. Frontier control alone cannot substitute for control of multiplicity.

Lackenby’s July 2026 preprint develops a hierarchy algorithm and efficient certificate operations [4]. Its introduction identifies speed-up as future work, and its quantitative analysis leaves a growth

parameter uncontrolled. The February 2021 Oxford slides announce  $n^{O(\log n)}$  [5]. The November 2024 Tsinghua abstract states the different announced bound

$$2^{O((\log n)^3)} = n^{O((\log n)^2)} \quad [6].$$

We distinguish that bound from the stronger  $n^{O(\log n)}$  target. Both are quasi-polynomial; they have different logarithmic exponents. We do not infer the reason for this difference from the announcements. The existence of an announcement does not make a local implementation quasi-polynomial without its missing quantitative and constructive steps.

## 1.2 A useful connection with openai/math

The second requested repository was inspected at revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its manuscript *Parabolic Intersections in Artin Groups*, dated 23 September 2026, develops finite-amplitude layer operations and explicit homotopies under tensoring [7]. The methodological connection is precise: bounded future degree movement makes some present layers irrelevant to a requested output. Section 3 proves the required version directly in the Khovanov setting. No theorem about arbitrary Artin-group parabolics is assumed by this recognizer.

## 2 The computational and algebraic model

An input is a validated one-component classical PD diagram with  $n$  crossings. Each crossing is a cyclic four-tuple; every edge label occurs twice. The code checks one-component connectivity and that the rotation system is spherical. After normalization the edge labels are dense integers, so the combinatorial input has  $O(n \log(n+2))$  bits. The empty PD represents one crossing-free circle.

A scan adds one crossing at a time. Its two resolutions have raw homological degrees 0 and 1. A resolved partial diagram is a matching of the current frontier, together with closed circles. Delooping replaces each circle by two summands over  $\mathbb{F}_2$ . If the frontier has  $w_i = 2k_i$  endpoints, a matching has  $k_i$  arcs. The scanner stores multiple copies of a matching in distinct chain degrees or with different differential incidences. These multiplicities are ordinary nonnegative integers, even though matrix coefficients lie in  $\mathbb{F}_2$ .

For two matchings  $a, b$ , let  $c(a, b)$  be the number of circles formed by gluing them along their frontier. The corresponding morphism space has  $2^{c(a, b)}$  dotted basis elements. An integer bit mask encodes their  $\mathbb{F}_2$  coefficients: bit position  $S$  records the monomial with a dot on each circle indexed by a bit of  $S$ . This integer can have as many as  $2^{w_i/2}$  bits. Treating it as a unit-cost scalar would hide an exponential dependence on  $w_i$ .

The quantity reported by the full backend is ungraded in the quantum direction and graded in raw homological degree. For a knot, the total unreduced  $\mathbb{F}_2$  rank is twice its reduced rank [9, Corollary 3.2.C]. The reduced-rank-one criterion detects the unknot: the characteristic-zero detection theorem [8] and the universal coefficient theorem imply that reduced  $\mathbb{F}_2$  rank one forces the characteristic-zero rank to be one; the unknot itself has that rank. Thus the existing full fallback has an exact mathematical certificate criterion. The partial window will use only the necessary homology of the unknot, so its agreement is inconclusive.

**Definition 2.1** (Parameters used below). Let  $w = \max_i w_i$  be the executed frontier size,  $s$  the number of Seifert circles,  $\nu$  the number of negative oriented PD crossings, and  $\mu = \min\{\nu, n - \nu\}$ . A normalized homological radius  $d$  asks for degrees  $[-d, d]$ . The raw target is  $[\nu - d, \nu + d]$ . For

the separate terminal algorithm,  $s_P$  denotes its number of darts; it is unrelated to the Seifert-circle parameter  $s$ .

### 3 Exact homology in a moving degree window

The next optimization changes which chain groups are constructed. It is useful when the decision procedure needs a prescribed part of Khovanov homology, and admits an unconditional complexity analysis for the delivered implementation. It does not assert that a bounded part of homology always detects a nontrivial knot.

Throughout this section the coefficient field is  $\mathbb{F}_2$ . The quantum grading is forgotten, but the raw cube degree is retained. A crossing complex is supported in raw degrees  $0, 1$ ; delooping changes no homological degree. If  $\nu$  denotes the number of negative crossings in `Diagram.signs()`, normalized degree  $h$  corresponds to raw degree  $h + \nu$ . This convention must be read from the oriented PD diagram: `Diagram.from_braid(2, [1])` has  $\nu = 1$  and raw homology in degree 1, while `Diagram.from_braid(2, [-1])` has  $\nu = 0$  and raw homology in degree 0. Both closures are unknots. Counting negative integers in a braid word instead would reverse this convention.

#### 3.1 The support lemma and its guard degrees

For a cochain complex  $C$ , write  $C[L, U]$  for its brutal interval truncation: retain  $C^j$  for  $L \leq j \leq U$ , retain the differentials between these groups, and set every other group and differential to zero. This is a complex, but its homology at  $L$  and  $U$  need not equal that of  $C$ .

**Lemma 3.1** (Support under a bounded continuation). *Suppose a remaining continuation is a composite of additive planar extension operations whose total raw homological support is contained in  $[u, v]$ , followed by closure and the Khovanov functor. To preserve final homology in raw degrees  $[a, b]$ , it suffices to replace the current complex  $C$  by*

$$C[a - v - 1, b - u + 1].$$

*This replacement gives exactly the same final chain groups and differentials in the three-degree neighborhood  $[a - 1, b + 1]$ .*

*Proof.* Write the total continuation formally as  $Q$ . A summand contributing in final degree  $j$  comes from current degree  $p$  and continuation degree  $q$  with  $p + q = j$  and  $u \leq q \leq v$ . If  $a - 1 \leq j \leq b + 1$ , then

$$a - v - 1 \leq p \leq b - u + 1.$$

Consequently no removed current summand contributes to any of these final chain groups. The total differential is the sum of the current and continuation differentials, with the usual tensor signs. Its entries between retained final groups are therefore unchanged as well. Applying the additive closure and Khovanov functors preserves these equalities. Finally,  $H^j = \ker(d^j) / \text{im}(d^{j-1})$  depends only on the groups in degrees  $j - 1, j, j + 1$  and the two indicated maps.  $\square$

Both guards are necessary in the general algebraic statement. The two-term complex  $\mathbb{F}_2 \xrightarrow{1} \mathbb{F}_2$  in degrees  $a - 1, a$  has zero  $H^a$ , whereas discarding degree  $a - 1$  creates a false class. Placing the same complex in degrees  $b, b + 1$  shows the corresponding problem at the upper end. Keeping only the desired degrees is therefore unsound even after the knot has closed.

For  $n - i$  unprocessed crossings the continuation support is  $[0, n - i]$ . Define

$$I_i = [L_i, U_i] \cap [0, i], \quad L_i = a - (n - i) - 1, \quad U_i = b + 1. \quad (1)$$

The upper endpoint is fixed. The lower endpoint moves upward by one after every crossing, because progressively fewer future 1-resolutions remain available to reach the target.

**Theorem 3.2** (Repeated windowing with arbitrary unit cancellation). *Start from the empty partial diagram, and process crossings in any valid scan order. After each extension, retain only the summands in  $I_i$  and the differential entries whose two endpoints are retained. Delooping and any sequence of valid Gaussian unit cancellations may be performed at each stage. The resulting closed complex has the exact original Khovanov homology in every raw degree  $a \leq j \leq b$ .*

*The discarded summands may be excluded before allocation, provided the constructed objects and maps equal the brutal truncation of the full next extension.*

*Proof.* Compare each intermediate complex only after applying its actual remaining continuation. At the beginning this gives the ordinary complete Khovanov complex. One crossing extension merely factors that continuation into its first operation and its suffix, so it leaves the completed complex unchanged. Lemma 3.1 shows that clipping to  $I_i$  preserves the completed homology in  $[a, b]$ .

A Gaussian cancellation is a chain homotopy equivalence of the current complex. Every remaining planar extension is additive and the induced operation on total complexes preserves the equations defining a chain homotopy. Closure and the Khovanov functor do the same. Hence a cancellation also preserves the completed target homology. Induction proves the claim.

This argument does not require the artificial endpoint homology of a truncation to be the original endpoint homology. Its possible effects are already outside the final target and its adjacent differential groups. Nor does it require the same pivots as an untruncated run. Pre-allocation pruning realizes the same truncated complex by omitting objects and maps that the support lemma would immediately delete.  $\square$

**Operational form.** The following sequence specifies the mathematical operation independently of the pivot heuristic:

1. Fix the full validated crossing order and target raw interval  $[a, b]$ .
2. At step  $i$ , compute the band  $I_i$  using the original total  $n$ .
3. For each current object of degree  $h$  and each smoothing  $\epsilon \in \{0, 1\}$ , allocate its delooped children only if  $h + \epsilon \in I_i$ .
4. Construct the crossing saddles and transferred old differentials only when both endpoint children were retained.
5. Cancel invertible entries, then continue to the next crossing.
6. At closure compute homology using both adjacent differentials, and return only the degrees in  $[a, b]$ .

The frontier itself belongs to the processed subdiagram. It must advance even when the retained algebraic complex is empty. A resource cap is a cap on the retained allocation; exhaustion gives an incomplete computation, never a homology value.

**Lemma 3.3** (Removing isolated guards). *After reduction at a prefix with  $r$  crossings remaining, an isolated object of raw degree  $h$  may be removed whenever  $[h, h + r] \cap [a, b] = \emptyset$ . A nonisolated guard cannot be removed by this test.*

*Proof.* An object with no incoming or outgoing nonzero differential entry is an independent direct summand. Completing this summand through the remaining crossings gives a complex supported in  $[h, h + r]$ . It therefore contributes zero homology to the target band. Direct sums commute with the continuation, so deleting the summand is valid. For a nonisolated object an incident differential may create or kill a target class, as the two-term examples above demonstrate.  $\square$

With the one-degree guards in (1), a differential component whose entire degree support misses  $[a - r, b]$  lies in one guard degree. Since differentials change raw degree by one, such components are already isolated objects. Thus this particular additional pruning does not require a general graph search.

### 3.2 An unconditional bound for the implemented scanner

Let  $s_0(D)$  and  $s_1(D)$  denote the numbers of circles in the complete all-0 and all-1 smoothings of a nonempty diagram  $D$ . These quantities are computed by pairing the appropriate crossing slots and counting components. Let  $w$  be the largest number of frontier endpoints in the chosen order. No assumption that these endpoints lie on one boundary circle is made.

For  $0 \leq k \leq n$ , put

$$Q(n, k) = \sum_{j=0}^k 2^j \binom{n}{j}, \quad B_0(D; a, b) = 2^{s_0(D)} Q(n, \min\{n, b + 1\}). \quad (2)$$

Targets are first intersected with  $[0, n]$ ; an empty intersection has zero homology. The separate input convention that an empty PD represents one circle is handled directly by the implementation.

**Lemma 3.4** (Circle and object counts). *Every raw cube state of a prefix with  $j$  1-resolutions has at most  $s_0(D) + j$  already closed circles. At every stage the number of retained physical summands in the window scanner, including the allocation before Gaussian reduction, is at most  $B_0(D; a, b)$ .*

*Proof.* Complete the prefix state with 0-resolutions at the remaining crossings. Every circle already closed in the prefix remains a distinct circle of that completion. The completed state differs from the all-0 state at  $j$  crossings. Changing one smoothing is a surgery on two arcs and changes the total circle count by at most one. The completed state therefore has at most  $s_0(D) + j$  circles.

A state with  $c$  already closed circles contributes  $2^c$  summands after delooping. There are  $\binom{i}{j}$  states of degree  $j$  at prefix length  $i$ . Extension and delooping enumerate the usual cube objects; Gaussian cancellation and truncation only remove objects. More formally, surviving objects can be assigned distinct original cube-basis labels, and extending these labels and adding new circle labels preserves this assignment. Consequently the degree- $j$  allocation has size at most  $2^{s_0(D)+j} \binom{i}{j}$ . Every retained degree satisfies  $j \leq b + 1$ . Summing and using  $\binom{i}{j} \leq \binom{n}{j}$  proves the bound.  $\square$

One can keep a more informative prefix bound. If  $z_i$  is the number of closed circles in the all-0 smoothing of the processed prefix, then the same local surgery argument gives

$$\#\text{objects after allocation at } i \leq \sum_{j \in I_i} 2^{z_i+j} \binom{i}{j}. \quad (3)$$

This explicitly exhibits the saving from the moving lower boundary. Since  $z_i \leq s_0(D)$ , it implies the uniform bound above.

**Theorem 3.5** (Bit complexity with unrestricted scan frontiers). *For the delivered  $\mathbb{F}_2$  dotted-cobordism representation there is an absolute constant  $c$  such that the uncapped window computation takes at most*

$$2^{c(w+1)}(n+1)^c(B_0(D; a, b) + 1)^c \quad (4)$$

*bit operations, after the normalized PD diagram and order are supplied. In particular, if  $k = \min\{n, b+1\}$ , its time is*

$$2^{O(w+s_0(D)+k)}(n+1)^{O(k+1)}. \quad (5)$$

*The implied constants do not depend on the knot, the order, or  $k$ .*

*Proof.* Two matchings on at most  $w$  endpoints have at most  $w/2$  circle components in their union. The implemented dot-mask basis of the corresponding morphism space therefore has at most  $2^{w/2}$  elements. A morphism is stored as at most that many bits. The compiled gluing plans use polynomially many boundary records; enumerating their monomials, composing maps and evaluating neck-cutting sums takes  $2^{O(w)}$  bit operations. Adding a crossing introduces only four local slots, which changes this bound by a constant factor in the exponent.

With  $B = B_0(D; a, b)$  objects, there are at most  $B^2$  differential entries. At most  $B/2$  unit cancellations occur in a stage; each cancellation examines at most  $B^2$  pairs of incident entries. Bookkeeping, the lazy pivot heap, stage-local caches and equality checks add only polynomial factors in  $n, B, w$ . The implementation uses  $\mathbb{F}_2$ , so coefficient growth contributes no additional integer-height parameter. There are  $n$  stages. Lemma 3.4 now gives (4). Finally,

$$Q(n, k) \leq (k+1)2^k(n+1)^k,$$

which yields (5).

This count never bounds the number of all possible matchings by a Catalan number. It bounds the matchings actually attached to retained cube summands, and then bounds the algebra attached to each pair. Thus the frontier issue for arbitrary crossing orders does not enter the proof.  $\square$

### 3.3 Choosing the less expensive mirror

Mirroring swaps the two smoothings. Over a field, the closed Khovanov complex of the mirror is dual to the original complex, with raw degrees reversed:

$$H_{\text{raw}}^j(D^!; \mathbb{F}_2) \cong H_{\text{raw}}^{n-j}(D; \mathbb{F}_2)^*.$$

The underlying crossing graph, and hence the frontier size in a fixed crossing order, is unchanged. Also  $s_0(D^!) = s_1(D)$ . Define

$$B_1(D; a, b) = 2^{s_1(D)} Q(n, \min\{n, n-a+1\}). \quad (6)$$

**Corollary 3.6** (A computable choice of direction). *Exact raw homology in  $[a, b]$  can be computed with the bound (4) after replacing  $B_0$  by  $\min\{B_0, B_1\}$ . If  $B_1 < B_0$ , run on the mirror with target  $[n-b, n-a]$ , then reverse the returned degree labels.*

*Proof.* Apply Theorem 3.5 to either  $D$  or  $D^!$ , using the displayed duality. The all-state circle counts are found by finite component calculations. Both  $Q$  values have  $O(n)$  bits, since  $Q(n, k) \leq 3^n$ , so comparing the two explicit bounds has polynomial cost.  $\square$

This choice accounts for the actual circle counts. Merely choosing the smaller extreme depth can miss a large difference between  $s_0$  and  $s_1$ . The choice minimizes this proved upper bound; it does not promise to predict the faster observed run on every input.

Let  $s$  be the number of Seifert circles, and let  $\mu = \min\{\nu, n - \nu\}$ . The oriented smoothing differs from the all-0 smoothing at exactly  $\nu$  crossings and from the all-1 smoothing at exactly  $n - \nu$  crossings in the stated PD convention. The same one-surgery circle bound gives

$$s_0 \leq s + \nu, \quad s_1 \leq s + (n - \nu).$$

For normalized target degrees  $[-d, d]$ , one may therefore choose a mirror and obtain the following useful uniform consequence.

**Corollary 3.7** (Minority-sign and Seifert-circle parameters). *For  $d \geq 0$ , exact Khovanov homology in normalized degrees  $[-d, d]$  can be computed by the window scanner in*

$$2^{O(w+s+\mu+d)}(n+1)^{O(\mu+d+1)} \quad (7)$$

*bit operations. In particular this is quasi-polynomial when  $w + s = O(\log^2(n+2))$  and  $\mu + d = O(\log(n+2))$ .*

*For a knot given as the closure of a  $t$ -strand braid scanned in braid order,  $s = t$  and  $w \leq 2t$ . The bound becomes*

$$2^{O(t+\mu+d)}(n+1)^{O(\mu+d+1)}.$$

*It is polynomial for fixed  $t, \mu, d$ , and quasi-polynomial for  $t = O(\log^2(n+2))$  and  $\mu + d = O(\log(n+2))$ .*

*Proof.* On the side with  $\mu$  negative PD crossings, the requested raw upper degree is at most  $\mu + d$ , and its upper guard is at most  $\mu + d + 1$ . Insert  $s_0 \leq s + \mu$  in Theorem 3.5. The logarithm of the resulting time is  $O(w + s + \mu + d + (\mu + d + 1) \log(n + 2))$ , proving the stated regimes.

Oriented smoothing of a braid closure yields its  $t$  Seifert circles. In braid order the processed prefix has at most one top and one current end per strand, so its frontier has at most  $2t$  endpoints. Substitution gives the braid assertions. The sign convention for braid-word generators does not affect  $\mu$ , although it does affect which mirror is selected.  $\square$

### 3.4 Scope and comparison with existing extremal algorithms

An exact band can reject an unknot whenever it differs from the unknot's homology in that band. In particular, excess dimension in normalized degree zero, or nonzero homology in a computed nonzero degree, is a sound certificate of nontriviality. An agreeing band remains inconclusive. For example, the trefoil has the unknot's dimension in degree zero and has additional homology elsewhere. A complete recognizer must use a fallback, eventually cover every degree, or prove a separate support or witness theorem for the input class.

Kelomäki–Schütz give an extremal-degree algorithm based on special “nice” scanning sequences and local cancellations [3]. Their Lemma 5.5 bounds intermediate degree- $j$  summands by  $\binom{i}{j}$ , with a factor two at closure; their finite-field Algorithm 3 permits exhaustive Gaussian cancellation. This

can remove the all-state circle factor in an appropriate geometric assembly. It is a distinct guarantee from the elementary object bound proved here. The current implementation accepts arbitrary valid crossing orders, and no claim that every such order satisfies their nice-scan hypotheses is made.

The cited work also proves polynomial-time computation of full homology for three-braid closures by specialized structure, despite exponential explicit-basis examples. Neither that specialized engine nor its general fixed-strand extension is supplied by degree clipping. Our contribution here is an exact moving interval implemented before allocation, together with a uniform, independently proved bound for that implementation.

**Relation to openai/math.** The source of *Parabolic intersections in Artin groups* in openai/math uses the principle that later functors with bounded level amplitude cannot bring discarded layers back to an extreme level; see its lemma “Composition at the bottom” in `build/sections/03_layers.tex` [7]. Its categorical-action section also retains explicit chain homotopies before tensoring. These are useful methodological parallels. The Artin-module levels are not the raw Khovanov degrees used here, and the present proof imports no theorem about Artin groups or their parabolic subgroups. Lemma 3.1 supplies the required statement directly in the scanner’s category. The reviewed repository revision is `adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

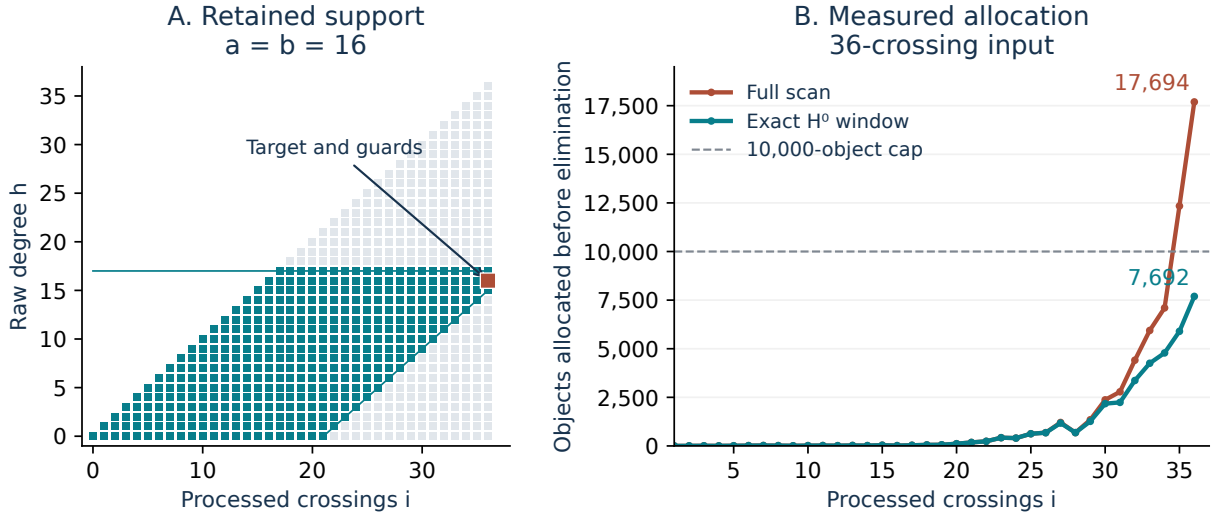


Figure 1: Left: the exact moving support window for  $n = 36$  and target raw degree 16. The guard degrees retain the incident differentials. Right: actual pre-elimination allocations for the supplied stress diagram under the same crossing order. The 10,000-object ceiling is crossed by the full scan and respected by the window scan.

## 4 A finite perturbation method for exact minimal complexes

The window scanner reduces the portion of a complex that must be constructed. A complementary question is how to reduce a complex once it exists. This section separates scalar linear algebra from the remaining cobordism calculation and derives a finite homological perturbation algorithm. The construction uses the established perturbation lemma [10]; the point here is an explicit nilpotence

bound for the implemented algebra, an exact multiplicity profile, and a checked implementation of the resulting maps.

All objects below have the same current frontier with  $w = 2k$  endpoints. They are loopless matchings, with raw homological placements retained and the quantum grading forgotten. The grading introduced next is an intrinsic grading of the morphism algebra. It is not the raw homological degree and does not require the program to reconstruct the forgotten quantum shifts.

#### 4.1 Positive grading without a Catalan frontier assumption

For two matchings  $a, b$ , let  $c(a, b)$  be the number of circle components in the union of  $a$  and a reversed copy of  $b$ . The implemented morphism basis consists of one canonical disk on each such circle, decorated with zero or one dot. This is the familiar dotted-cobordism and arc-algebra description[2, 11]. A basis element carrying  $d$  dots receives intrinsic degree

$$\ell(a, b; d) = k - c(a, b) + 2d. \quad (8)$$

The relevant matching family is the one occurring in the executed scan. Its endpoints need not lie on one noose, and its matchings need not be noncrossing with respect to their numerical labels. Nevertheless the spherical projection has a checkerboard face coloring. Orient each edge so that the black face is on its left. At each crossing the four edge directions alternate between incoming and outgoing, and either smoothing joins incoming to outgoing. At a fixed cut this gives a fixed division of the frontier into  $k$  positive and  $k$  negative endpoints; every smoothing matching joins opposite classes.

Thus the abstract disk gluings used below are oriented. This is the condition needed for the Euler-characteristic calculation, and it does not require a Catalan count. One should not instead declare every unconstrained perfect matching on a labeled set admissible: for four endpoints the three distinct pairings together need not admit the required common endpoint orientations. The present argument uses matchings actually furnished by a spherical diagram.

**Lemma 4.1** (Homogeneous composition and degree range). *Every nonzero basis morphism has intrinsic degree in  $[0, w]$ . Composition adds intrinsic degrees. Degree zero consists precisely of scalar identities between equal matchings.*

*Proof.* Since  $1 \leq c(a, b) \leq k$  for a nonempty frontier and  $0 \leq d \leq c(a, b)$ ,

$$0 \leq k - c(a, b) + 2d \leq k + c(a, b) \leq 2k = w.$$

At the empty frontier only the empty object and its scalar endomorphisms remain, so the statement also holds for  $w = 0$ . Degree zero requires  $c(a, b) = k$  and  $d = 0$ . The union has  $k$  two-edge circles exactly when  $a = b$ , and the undotted canonical cobordism is then the identity.

For composition from  $a$  through  $b$  to  $c$ , abbreviate  $c_{ab} = c(a, b)$ ,  $c_{bc} = c(b, c)$  and  $c_{ac} = c(a, c)$ . Before reduction, the composite surface has Euler characteristic

$$\chi = c_{ab} + c_{bc} - k, \quad (9)$$

because  $k$  intervals of the middle matching are glued. Let  $d_{\text{in}}$  be the total input dot count. Replacing this surface by its canonical output disks through neck cutting raises Euler characteristic by two for each additional dot. Accordingly every nonzero output monomial satisfies

$$2d_{\text{out}} = 2d_{\text{in}} + c_{ac} - \chi. \quad (10)$$

The factor two in this identity is essential. A positive-genus component contributes zero over  $\mathbb{F}_2$  after handle reduction; double dots and undotted spheres also vanish, while a dotted sphere evaluates to one. These relations preserve the same degree balance whenever a term remains. Therefore

$$\begin{aligned}\ell_{\text{out}} &= k - c_{ac} + 2d_{\text{out}} \\ &= k - \chi + 2d_{\text{in}} \\ &= 2k - c_{ab} - c_{bc} + 2d_{\text{in}} = \ell_{\text{first}} + \ell_{\text{second}}.\end{aligned}$$

The calculation concerns abstract oriented surfaces, so numerical endpoint order and the number of boundary components of the scanned region do not enter it.  $\square$

Let  $J$  be the ideal of positive intrinsic-degree morphisms. For a finite set  $\mathcal{M}$  of distinct executed matchings, the basic algebra is  $A = \text{End}(\bigoplus_{a \in \mathcal{M}} a)$ .

**Corollary 4.2** (Uniform nilpotence). *One has*

$$J^{w+1} = 0, \quad A/J \cong \prod_{a \in \mathcal{M}} \mathbb{F}_2.$$

*The same nilpotence exponent applies to matrices of arbitrary finite size over this matching category. In the additive closure, an object with  $m_a$  copies of each matching has endomorphism quotient  $\prod_a \text{Mat}_{m_a}(\mathbb{F}_2)$ , rather than a product of one-dimensional fields. The ideal  $J$  is the radical of these finite-dimensional algebras.*

*Proof.* A product of  $w + 1$  positive-degree basis terms would have degree at least  $w + 1$ , where all morphism spaces vanish by Lemma 4.1. Expand sums and matrix products into composable paths to obtain the same conclusion for arbitrary matrix sizes. The degree-zero description gives the displayed quotients. A nilpotent ideal lies in the radical, while a semisimple quotient forces the reverse containment here.  $\square$

## 4.2 The scalar differential gives the exact remaining dimensions

Let  $(C, d)$  be a finite complex over this category. Extract  $d_0$  by keeping only the undotted constant coefficient between equal matchings; put  $\delta = d + d_0$ . Every entry of  $\delta$  lies in  $J$ . Projection to degree zero is compatible with composition, so  $d^2 = 0$  implies  $d_0^2 = 0$ .

For each matching  $a$  and raw degree  $h$ , let  $V_{a,h}$  be the vector space of its copies in  $C^h$ . The scalar differential forms independent binary complexes

$$\cdots \longrightarrow V_{a,h-1} \xrightarrow{d_{0,a,h-1}} V_{a,h} \xrightarrow{d_{0,a,h}} V_{a,h+1} \longrightarrow \cdots$$

Define

$$m_{a,h} = \dim V_{a,h} - \text{rank}(d_{0,a,h-1}) - \text{rank}(d_{0,a,h}). \quad (11)$$

**Theorem 4.3** (Exact minimal multiplicity profile). *There exists a complex homotopy equivalent to  $(C, d)$  whose differential lies entirely in  $J$  and which has exactly  $m_{a,h}$  copies of matching  $a$  in degree  $h$ . Every such minimal model has the same multiplicities. Thus the number of remaining objects can be computed by binary linear algebra before carrying out the nonlinear cobordism correction.*

*Construction of the scalar contraction.* In every binary complex choose decompositions

$$V_{a,h} = B_{a,h} \oplus H_{a,h} \oplus L_{a,h}, \quad B_{a,h} = \text{im } d_{0,a,h-1},$$

where  $B_{a,h} \oplus H_{a,h} = \ker d_{0,a,h}$  and  $d_0 : L_{a,h} \rightarrow B_{a,h+1}$  is an isomorphism. Let  $H = \bigoplus_{a,h} a^{m_{a,h}}[h]$ , using  $[h]$  here only to record a raw placement. Let  $i : H \rightarrow C$  include the chosen homology representatives,  $p : C \rightarrow H$  project along  $B \oplus L$ , and  $h : C \rightarrow C$  take each boundary back to its chosen preimage in  $L$ , vanishing on  $H \oplus L$ .

All these maps have scalar identity entries between equal matchings. They satisfy

$$pi = 1, \quad d_0i = 0, \quad pd_0 = 0, \quad d_0h + hd_0 = 1 + ip, \quad hi = 0, \quad ph = 0, \quad h^2 = 0. \quad (12)$$

The next theorem lifts this contraction to the full  $d$  without changing  $H$  as an object. Its resulting differential lies in  $J$ . Uniqueness is proved below.  $\square$

The program obtains  $B$  from independent original image columns and  $L$  from their corresponding source vectors. It extends  $B$  to a basis of the kernel to choose  $H$ , and solves for coordinates in  $B \oplus H \oplus L$ . This provides the actual maps in (12), not only the ranks in (11).

### 4.3 A finite homological perturbation formula

Composition is written in the conventional order:  $fg$  means first  $g$ , then  $f$ . The code's composition helper takes its arguments in execution order: the second map acts after the first.

**Theorem 4.4** (Finite lifted contraction). *Assume (12), let  $d = d_0 + \delta$  with  $d^2 = 0$ , and suppose the entries of  $\delta$  lie in  $J$ . Define*

$$F = \sum_{j=0}^w (h\delta)^j = (1 + h\delta)^{-1}, \quad G = \sum_{j=0}^w (\delta h)^j = (1 + \delta h)^{-1}, \quad (13)$$

$$\begin{aligned} i' &= Fi, & p' &= pG, & h' &= Fh = hG, \\ D &= p\delta Fi = pG\delta i. \end{aligned} \quad (14)$$

*These are finite sums, and  $(H, D)$  is a minimal complex with a strong deformation retraction from  $(C, d)$ . Explicitly,*

$$D^2 = 0, \quad di' = i'D, \quad p'd = Dp', \quad p'i' = 1, \quad (15)$$

$$dh' + h'd = 1 + i'p', \quad h'i' = 0, \quad p'h' = 0, \quad (h')^2 = 0. \quad (16)$$

*The number of perturbation factors needed is at most  $w$ , independent of the number of objects or the size of their scalar blocks.*

*Proof.* The maps  $i, p, h$  have intrinsic degree zero. Therefore  $h\delta$  and  $\delta h$  have entries in  $J$ , and Corollary 4.2 proves the inverse identities in (13). These are the standard perturbation formulas [10, Sections 2–3]; a direct verification in characteristic two is short enough to record.

Put  $e = ip$  and

$$A = \delta F = G\delta.$$

The finite series imply

$$A = \delta + \delta hA = \delta + Ah\delta, \quad F = 1 + hA, \quad G = 1 + Ah. \quad (17)$$

There is also the identity

$$d_0A + Ad_0 = AeA. \quad (18)$$

To verify it, multiply its left side by  $1 + \delta h$  on the left and  $1 + h\delta$  on the right. Using  $AF^{-1} = \delta$  and  $G^{-1}A = \delta$  gives

$$\begin{aligned} (1 + \delta h)(d_0A + Ad_0)(1 + h\delta) \\ = d_0\delta + \delta d_0 + \delta(hd_0 + d_0h)\delta \\ = \delta^2 + \delta(1 + e)\delta = \delta e\delta. \end{aligned}$$

Here  $d^2 = 0$  gives  $d_0\delta + \delta d_0 = \delta^2$  over  $\mathbb{F}_2$ . Multiplying by  $G$  and  $F$  proves (18).

Now  $D = pAi$ , so

$$D^2 = pAeAi = p(d_0A + Ad_0)i = 0.$$

Since  $i' = i + hAi$ , equations (12) and (17) give

$$di' = eAi + hd_0Ai = eAi + hAeAi = i'D.$$

The same calculation on the other side gives  $p'd = Dp'$ . Expanding  $p'i' = (p + pAh)(i + hAi)$  leaves only  $pi = 1$ , by the three side conditions  $hi = ph = h^2 = 0$ .

Finally  $h' = h + hAh$ . Expand  $dh' + h'd$  and use  $A = \delta + \delta hA = \delta + Ah\delta$  to cancel the terms containing a single  $\delta$ . What remains is

$$\begin{aligned} dh' + h'd &= 1 + e + eAh + hAe + h(d_0A + Ad_0)h \\ &= 1 + e + eAh + hAe + hAeAh = 1 + i'p'. \end{aligned}$$

Expanding  $h'i'$ ,  $p'h'$  and  $(h')^2$  similarly makes every term contain  $hi$ ,  $ph$  or  $h^2$ , so each vanishes. Each term of  $D = p\delta Fi$  contains a positive-degree factor  $\delta$ , hence  $D$  lies in  $J$  and is minimal.  $\square$

**Lemma 4.5** (Uniqueness of minimal models). *Two finite complexes with differentials in  $J$  that are homotopy equivalent are chain isomorphic. Their multiplicities agree matching by matching and raw degree by raw degree. In particular, the multiplicities in Theorem 4.3 agree with any exhaustive unit cancellation result.*

*Proof.* Reduce a homotopy equivalence and its inverse modulo  $J$ . Both reduced differentials are zero, so the homotopy equations say that the reduced term maps are inverse in each degree. Thus the binary multiplicity spaces agree. Each lifted composite differs from the identity by a matrix in  $J$ . Such a matrix is inverted by a finite geometric series of length at most  $w + 1$ . Consequently each original term map is invertible; its inverse commutes with the differential because the term map does. This gives a chain isomorphism. The assertion concerns multiplicities and isomorphism type, not a canonical displayed basis or identical matrices.  $\square$

#### 4.4 What the implementation checks

The research module `perturbation.py` first constructs the scalar contraction, then computes  $i'$  by repeatedly applying  $h\delta$  to its current term. It obtains  $D$  from  $p\delta i'$ . With certification enabled it also constructs  $p', h'$  and verifies all identities in (15)–(16), together with the original  $d^2 = 0$ , over the same exact cobordism algebra used by the scanner. It checks the finite-series termination and the absence of scalar units in the output differential.

The corrected maps are stronger evidence than an agreement of final ranks: they give the explicit homotopy equivalence needed for continuing the scan. Because planar extension preserves homotopies, an installed model can be extended through subsequent crossings. The test suite compares its matching-and-degree multiplicities with exhaustive scalar cancellation at intermediate prefixes, checks final homology independently, and confirms that a corrupted contraction map is rejected. An additional independent grading audit examined all bipartite matching monomial products through six endpoints, including the non-Catalan six-endpoint matching: all 3533 products and 1576 nonzero output monomials obeyed Lemma 4.1.

#### 4.5 Cost, useful regimes, and remaining obstruction

Write  $N$  for the input object count at one stage and  $H = \sum_{a,h} m_{a,h}$  for the output count. There are at most  $2^{w/2}$  dot-mask coefficients in one morphism. Scalar contraction uses binary matrices with at most  $N$  rows and columns and therefore has polynomial bit cost, with a standard cubic bound sufficient here. The inclusion series performs at most  $w$  iterations on matrices of size at most  $N$  by  $H$ . A conservative bound for the reduction without optional certificates is

$$O(N^3 + (w+1)N^2(H+1)2^{O(w)}). \quad (19)$$

Computing and verifying all contraction maps has the coarser bound

$$O((w+1)(N+1)^3 2^{O(w)}).$$

These estimates allow dense maps; sparsity can help but is not assumed. They follow by counting ordinary matrix-composition paths and charging  $2^{O(w)}$  for a cobordism product and its bit operations. Cache storage and dictionary bookkeeping remain polynomial in  $N$  times  $2^{O(w)}$ .

The attraction is most pronounced when the scalar differential has large rank, so  $H \ll N$ . In the extreme  $H = 0$ , scalar acyclicity already implies full contractibility: the corrected inclusion and reduced differential have empty source. If an explicit contracting homotopy is requested, its construction still has a cost. When scalar cancellation is sparse and cheap, assembling a global binary decomposition and dense corrected maps can instead add overhead. Thus no pointwise or asymptotic speed advantage over the existing scalar scanner is claimed. The experiment section reports the measured regimes, including the distinction between natural knot complexes and synthetic algebraic blocks.

This method does not control  $N$  or  $H$  by frontier width alone. It computes the minimal multiplicity profile of an already represented complex, and an explicit output must still list its  $H$  objects. Known fixed-braid-width exponential examples therefore remain an obstruction to a universal explicit-basis complexity theorem. The finite perturbation bound supplies a precise local reduction cost and a certificate; combining it with window bounds or a separate structural compression theorem is the remaining route to stronger global guarantees.

### 5 A deterministic near-linear terminal boundary-pattern algorithm

The previous terminal implementation in report 06 tests all sets of at most three primal edges and traverses the remaining graph for each set. Its  $O(E^4)$  worst-case bound can be improved to  $O(s_P \log(s_P + 2))$  deterministic word-RAM operations, where  $s_P = 2E$  is the number of darts. The improvement is an implemented graph algorithm with an independently checked negative certificate.

It does not require a generic four-connectivity routine. This is an algorithmic consequence of the established theory of essential boundary patterns, rather than a new topological characterization.

## 5.1 Input and topological precondition

The ambient manifold is a known 3-ball  $B$ . Its boundary pattern  $P$  is a disjoint union of embedded trivalent multigraphs and vertex-free circles. The input contains a cyclic triple at each graph vertex. Dart  $2e$  is paired with dart  $2e + 1$  by the fixed-point-free involution  $\alpha(d) = d \text{ xor } 1$ . Write  $\sigma$  for the permutation whose cycles are the supplied triples. The face walks are the cycles of  $\phi = \sigma\alpha$ .

The implementation checks that the dart list is exactly a permutation of  $\{0, \dots, s_P - 1\}$ , that every vertex has degree three, and that  $V_i - E_i + F_i = 2$  for every connected graph component. Thus each connected rotation system has a spherical embedding. A rotation system does not specify the relative nesting of distinct graph components. This omission does not affect the essentiality decision: any pattern with two or more components is inessential, independently of their nesting.

The supplied assertion that the ambient manifold is a 3-ball remains a precondition. In particular, a spherical boundary does not by itself establish this precondition. The routine does not decide whether an arbitrary 3-manifold is a ball.

## 5.2 The dual triangle criterion

Recall that a pattern on a ball is essential when every transverse simple closed curve meeting  $P$  at most three times bounds a boundary disc whose intersection with  $P$  is empty, a single arc, or a single tripod. Lackenby's Proposition 2.6 [4] characterizes this property using the planar dual and four-connectivity. We use the following equivalent form.

**Theorem 5.1** (Terminal dual triangle criterion). *Let  $P$  be a nonempty connected trivalent graph embedded in a sphere, and let  $G$  be its embedded dual, with loops and multiple edges retained. Then the pattern is essential if and only if:*

1.  $G$  has no loops;
2.  $G$  has no two distinct edges with the same pair of endpoints;
3. every three-cycle of  $G$  is the boundary of a face of  $G$ .

*The complete patterns consisting of the empty set or one vertex-free circle are also essential. Every disconnected nonempty pattern is inessential.*

*Proof.* We use the elementary bond–cycle duality for connected sphere-embedded graphs: a minimal nonempty primal edge cut is dual to a simple dual cycle, and conversely. One way to see this is to thicken either side of a minimal edge cut and fill the complementary discs containing no vertices on the other side. Connectedness on both sides leaves one boundary curve, crossing exactly the cut edges. Its arcs through the primal faces give the dual cycle. Conversely, a simple dual cycle is a Jordan curve; its two sides give connected primal vertex sets and hence a minimal edge cut. Loops and pairs of parallel dual edges correspond to cuts of size one and two, respectively.

An edge cut of size one or two gives a violating boundary curve. A three-edge bond is harmless precisely when one side has a single primal vertex: the boundary disc then meets the trivalent graph in a tripod. In the dual, this is precisely the condition that its three-cycle is facial. Thus every obstruction in (1)–(3) gives a violating curve.

For completeness, suppose that (1) and (2) hold. Bond–cycle duality implies that the primal graph has no edge cut of size one or two. Let  $\gamma$  be a simple closed curve meeting the graph in  $k \leq 3$  points. If one side contains no primal vertex, its intersection with the graph is either empty or, when  $k = 2$ , a single arc; that side is allowed. Otherwise let  $U$  be the vertices on one side. The edge boundary  $\delta(U)$  has at most  $k$  edges. Since smaller edge cuts do not exist,  $|\delta(U)| = k = 3$ , and  $\gamma$  crosses each of these three edges exactly once. The induced graphs on  $U$  and on its complement are connected: two components on one side would each have at least three outgoing edges, exceeding the available three. Thus  $\delta(U)$  is a three-edge bond. By (3), its dual triangle is facial, so one side has a single primal vertex and is an allowed tripod disc. This proves essentiality.

The empty pattern and a sole circle satisfy the definition directly. When a pattern is disconnected, there is a separating curve disjoint from it with nonempty pattern on both sides. This is violating.  $\square$

The face occurrence convention matters at the smallest input. For  $G = K_3$ , its unique three-cycle bounds two different faces. The primal pattern is the theta graph, with two tripod sides, and is essential. The implementation records two facial occurrences and one distinct facial vertex triple. It accepts the triangle when at least one facial occurrence exists. The  $K_4$  exception is handled in exactly the same way, with four distinct facial triples.

### 5.3 Algorithm and complexity

After validation and the disconnected, empty, and circle cases, construct each dual edge as  $(f(d), f(\alpha(d)), e)$ , where  $f(d)$  is the face containing dart  $d$ . Sorting these triples detects loops and repeated endpoint pairs. Once the dual is simple, the three faces around each primal vertex form a facial triangle of the dual; retain these triples, sorted with multiplicity.

Every nonempty subgraph of a simple planar graph has a vertex of degree at most five. Delete such vertices successively. A queue implementation adds a vertex initially if its degree is at most five, or later when its degree first falls from six to five. Thus every vertex is queued once and every edge is examined a constant number of times.

Orient each edge from its earlier-deleted endpoint to its later-deleted endpoint. There are at most five forward neighbors at every vertex. For each unordered pair of forward neighbors, use binary search in the sorted edge list to decide whether the pair completes a triangle. A triangle is encountered exactly once, at its earliest-deleted vertex. Use a second binary search in the list of facial triples to decide whether the triangle is facial.

**Theorem 5.2** (Implemented terminal complexity). *For a valid input with  $s_P$  darts, the classifier uses  $O(s_P \log(s_P + 2))$  deterministic word-RAM operations and  $O(s_P)$  words of space. Its triangle scan examines at most  $10|V(G)|$  candidate neighbor pairs. All labels are dense integers of  $O(\log(s_P + 2))$  bits; a conservative charge of this many bit operations per word operation gives  $O(s_P \log^2(s_P + 2))$ . The separately supplied circle count also contributes its own binary input length.*

*Proof.* Validation, face traversal, dual construction, and degree peeling take linear time apart from sorting. All edge and face lists have  $O(s_P)$  entries, so sorting costs  $O(s_P \log(s_P + 2))$ . Each of the  $|V(G)|$  vertices contributes at most  $\binom{5}{2} = 10$  candidate pairs. Each candidate uses a constant number of binary searches and therefore  $O(\log(s_P + 2))$  comparisons. A returned negative certificate requires one additional primal traversal and sorting its two vertex lists, again within the stated bound. Every stored list has linear size.  $\square$

No hash-table assumption enters this bound. The implementation uses sorted edge and face arrays and a queue. The degree peeling does not search for a minimum-degree vertex; any available vertex of degree at most five suffices.

## 5.4 Certificates and independent verification

The three connected negative certificate kinds are a dual loop, a dual bigon, and a nonfacial dual triangle. A certificate records the cyclic list of dual vertices, the crossed primal edge labels, and the two primal vertex sets obtained by deleting these edges. A separate verifier checks all labels, traverses the primal graph after deleting the indicated edges, and confirms that the edge boundary of one side is exactly the indicated set. For three edges it also checks that both sides contain more than one vertex. Thus the verifier does not rerun the triangle enumeration.

These data determine a violating curve on the known boundary sphere: join consecutive crossings within their specified primal face. Pushing one boundary disc slightly into the ball gives a violating properly embedded disc. The disconnected certificate establishes existence of a zero-intersection violating curve; without relative nesting data it does not specify a particular geometric representative.

A terminal witness is not yet a normal compression disc in the original knot exterior. Transporting it backwards through a hierarchy still requires the cut surfaces, boundary identifications, and changes made during parallelity-bundle simplification. The present improvement removes repeated small-cut enumeration at the terminal stage; it does not supply those geometric operations.

## 5.5 Validation and reproducible comparison

The tests compare the new classifier and its certificates against the unmodified report-06 bond enumerator. The small-map test exhausts all  $15 + 10395 = 10410$  pairings of darts at two and four cyclically ordered trivalent vertices. Of these, 5628 have spherical graph components and 4782 are rejected as non-spherical. Every accepted input is checked against the old classifier; every returned connected negative certificate is also translated into the old verifier's primal-bond format and checked there.

Additional tests use random larger ribbon maps, sphere triangulations generated by stellar subdivisions and legal edge flips, relabeling and mirroring, malformed inputs, corrupted witnesses, all three obstruction types, the  $K_3$  double-face case, and an essential pattern with 60000 darts. The deterministic benchmark compares complete calls from a prepared JSON-like input through validation to the returned result. Fixture construction and external JSON reading/writing are excluded from both timings.

The full raw timing records, machine description, source provenance, and generator parameters are included in the accompanying artifacts. They measure this terminal subroutine and do not establish a running-time bound for a complete hierarchy-based unknot recognizer.

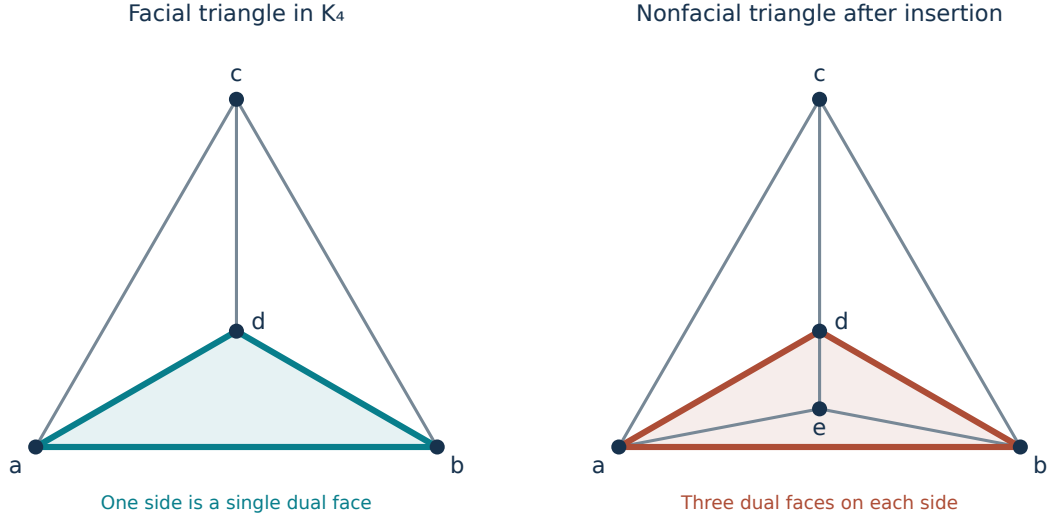


Figure 2: The dual criterion on explicit sphere triangulations. A facial triangle in  $K_4$  is harmless. Inserting a vertex into that face makes its old boundary a nonfacial triangle: the two sides contain three primal vertices each, producing a three-edge violating bond.

## 6 An output-size obstruction for compressed cuts

An efficient compressed cutting theorem must be translated into an explicit output contract. Merely storing normal coordinates in binary does not make an expanded component list efficient.

**Proposition 6.1** (Parallel-disc output lower bound). *There is a fixed one-tetrahedron triangulated 3-ball and a family of properly embedded orientable normal surfaces  $S_W$ , for integers  $W \geq 1$ , with normal-coordinate input length  $O(\log(W+1))$ , such that cutting along  $S_W$  produces  $W+1$  connected components. Between successive normal discs are  $W-1$  pairwise disjoint product slabs. Consequently an algorithm that explicitly lists all cut components, or all these individual product slabs, requires  $\Omega(W)$  output operations.*

*Proof.* Choose a vertex of one tetrahedron and take  $W$  pairwise disjoint, nested, normally isotopic triangles linking that vertex. Their normal vector has one coordinate  $W$  and six zero coordinates. Each triangle is a properly embedded disc. In a collar of one such disc, the successive triangles divide the collar into  $W-1$  copies of  $D^2 \times I$ . The two remaining regions, one towards the vertex and one away from it, are also balls. Therefore the cut manifold has  $W+1$  components. Writing one record for each component or slab has the asserted lower bound. Taking  $W = 2^k$  makes this exponential in the binary coordinate length.  $\square$

This does not contradict a compressed-cutting theorem. It shows that its concrete interface must support aggregated multiplicities, symbolic families, or a stated connectedness restriction. A suitable record for this example is one ordinary product type with binary multiplicity  $W-1$ , together with the two exceptional end regions. It is insufficient to advertise polynomial time in  $\log W$  while returning a Python list with one element per product region.

More generally, if  $S$  is a connected two-sided normal surface with normal vector  $v$ , then  $Wv$  is

represented by  $W$  parallel copies of  $S$ . With compatible collar markings,

$$M \setminus (WS) \cong (M \setminus S) \sqcup \coprod_{i=1}^{W-1} (S \times I).$$

Thus multiplicity extraction can be an exact preprocessing operation when connectedness and two-sidedness of the base surface are certified. Dividing coordinates by their greatest common divisor alone does not certify either condition. For a one-sided connected surface, doubling its normal vector describes the boundary of a regular neighborhood, which can be connected; blindly interpreting that factor of two as two disjoint copies would be incorrect.

In the July 2026 hierarchy paper, Theorem 14.4 (p. 51) gives the compressed cut information, whereas Proposition 14.2 (p. 50) adds essential-pattern and irreducibility hypotheses for the subsequent simplification. These are different contracts. In particular, an undecided unknot exterior does not supply an essential initial pattern as an available assumption. The representation lower bound above is an implementation obligation, not an assertion that the literature’s compressed theorem is false.

## 7 Quantitative accounting for a future hierarchy engine

The near-linear terminal test removes one repeated search from the hierarchy route. To obtain a general quasi-polynomial recognizer, one must also bound the construction and revision of the hierarchy, the size of its compressed states, and the cost of transporting a terminal witness. This section specifies a sufficient global accounting theorem. It does not assert that the geometric hypotheses have been established for the delivered implementation.

### 7.1 The precise status of the source bounds

The February 2021 slides cited here, the file `quasipolynomial-talk.pdf`, announce  $n^{O(\log n)}$  [5]. The November 2024 Tsinghua abstract announces

$$2^{O((\log n)^3)} = n^{O((\log n)^2)} \quad [6].$$

These are different stated bounds. Both are quasi-polynomial, and the sources reviewed here do not establish an explanation for the difference. The July 2026 preprint is a separate source: it gives a hierarchy algorithm and efficient certificate operations, while identifying further speed-up as future work [4].

For its iteration analysis, let  $L$  bound the length of every hierarchy that occurs, and let  $g$  bound every nonnegative complexity digit that is stored, including after a suffix is rebuilt. Proposition 9.1 gives

$$\#\{\text{executions of any fixed algorithmic step}\} \leq L(g+1)^L. \quad (20)$$

Proposition 10.2 gives the refined length bound  $L \leq 4c_q(\mathcal{H})$ , in terms of the initial handle structure’s  $q$ -complexity. For a linear-size initial structure of uniform type this is  $L = O(n)$  [4, Props. 9.1 and 10.2].

Equation (20) is not yet a bit-time estimate. Its digits concern pattern complexity, including intersections with the boundary pattern, rather than genus alone. A bound valid only for the first

chosen surfaces is insufficient when later replacements introduce new surfaces. Moreover, each counted step may act on a compressed object whose bit length must itself be controlled.

Even if the additional estimate  $g \leq (n + 2)^a$  were supplied, combining it only with  $L = O(n)$  would give the coarse iteration bound  $\exp(O(n \log(n + 2)))$ . This is not a quasi-polynomial upper bound. It is also not a lower bound on the algorithm: a sharper analysis may exploit correlations between successive choices or long stretches of forced operations. The distinction motivates the following elementary accounting result.

## 7.2 A rooted-tree bound that includes forced operations

All logarithms in this subsection are natural logarithms. A rooted tree represents *actual execution occurrences*: distinct visits are distinct nodes unless a separate memoization argument proves that they can be charged to one computation. At a node  $v$ , write  $b(v)$  for the number of its children. A node with one child describes a forced operation and contributes  $\log b(v) = 0$ . Leaves contribute no logarithm.

**Theorem 7.1** (Depth and cumulative branching). *Let  $T$  be a rooted, finitely branching tree of depth at most  $D$ , where depth counts edges. Suppose that every root-to-leaf path satisfies*

$$\sum_{\substack{v \text{ on the path} \\ b(v) > 0}} \log b(v) \leq B. \quad (21)$$

Then

$$|V(T)| \leq (D + 1)e^B. \quad (22)$$

*If all work associated with one node has bit cost at most  $C$ , including its assigned state construction, tests, certificate data, and transition work, the total assigned cost is at most  $C(D + 1)e^B$ .*

*Proof.* Finite branching and bounded depth imply that  $T$  is finite. Assign the root weight 1, and assign every child of an internal node  $v$  the weight of  $v$  divided by  $b(v)$ . Thus at every fixed depth the sum of the node weights is at most 1: passing to the next depth preserves the weight at internal nodes and drops the weight at leaves.

For a node  $u$ , its weight is

$$p(u) = \prod_{\substack{v \text{ a proper} \\ \text{ancestor of } u}} \frac{1}{b(v)}.$$

Extend its path to a leaf. Since every summand in (21) is nonnegative, the ancestor sum for  $u$  is at most  $B$ . Hence  $p(u) \geq e^{-B}$ . If there are  $N_j$  nodes at depth  $j$ , then  $N_j e^{-B} \leq 1$ , so  $N_j \leq e^B$ . Summing over  $j = 0, \dots, D$  proves (22). Summing the stated per-node costs proves the final assertion.  $\square$

The weights are a counting device. No randomized algorithm or probabilistic running-time assumption is involved. The same proof works with computable bounds  $\hat{b}(v) \geq b(v)$  in place of the actual positive child counts: distribute weight  $p(v)/\hat{b}(v)$  to each child, so that total weight can only decrease.

Unary chains explain why the depth term cannot be omitted. A chain of  $D + 1$  nodes has  $B = 0$  and attains the bound exactly. Conversely, a small bound on the length of each completed geometric hierarchy is not automatically a bound on  $D$ : rebuilding its suffix many times can create many execution occurrences. A tree interpretation must account for those occurrences explicitly.

**Corollary 7.2** (Explicit quasi-polynomial budgets). *Suppose a complete algorithm on  $n$ -crossing inputs admits the accounting of Theorem 7.1, uniformly over all its inputs and executions. If*

$$B(n) + \log(D(n) + 1) + \log(C(n) + 1) = O((\log(n + 2))^r)$$

*for a fixed  $r \geq 1$ , then its accounted bit time is  $\exp(O((\log(n + 2))^r))$ . In particular, polynomial  $C, D$  and  $B = O(\log^2(n + 2))$  imply  $n^{O(\log(n+2))}$  time; replacing that branching budget by  $O(\log^3(n + 2))$  gives  $n^{O((\log(n+2))^2)}$ .*

*Proof.* Take logarithms of the time bound in Theorem 7.1 and substitute the hypotheses. □

For example, polynomially many forced operations along a path are compatible with the stronger bound when at most  $O(\log n)$  nodes branch and each has at most  $n^{O(1)}$  children. Their cumulative branching is  $O(\log^2 n)$ . If polynomial branching occurs at  $O(\log^2 n)$  levels instead, the same arithmetic gives  $O(\log^3 n)$ . These are sufficient accounting regimes, not an explanation of the historical announcements.

This formulation permits the available choices to depend on the entire earlier history. It can therefore be more informative than multiplying independent worst-case caps at every hierarchy level. Nevertheless, the bound must hold on all explored paths, including failed choices and restarts. A bound on one successful certificate, or on the number of distinct unlabeled states, does not establish the hypotheses.

### 7.3 The geometric obligations behind the symbols

There are three distinct representation issues. First, a polynomial number of local handle types does not bound the number of handles. Second, large normal coordinates require binary arithmetic and multiplicity-aware output, as Proposition 6.1 demonstrates. Third, a compact object can still require a costly search to construct. The per-node bound  $C$  must include all three issues rather than treating surface production or witness transport as an oracle.

Theorem 14.4 of the 2026 paper supplies compressed cutting data in time polynomial in the handle count and logarithm of the extended surface weight. Proposition 14.2's subsequent simplification has additional irreducibility and essential-pattern hypotheses [4, Prop. 14.2 and Thm. 14.4]. The latter hypothesis cannot be silently supplied by an undecided knot exterior with empty initial pattern. A verifier may establish it from an appropriate certificate; a general producer must also handle the opposite case.

A concrete realization of Corollary 7.2 should therefore attach to every state an encoding-size bound, an explicit description of its admissible continuations, and a record of each operation used to transfer boundary curves. If a simplification is conditional on essentiality, its contract must state how that precondition is certified or how a compression witness is handled. All outer changes of Heegaard data and all reconstructions of discarded suffixes must receive a charge in the same execution accounting.

Efficient verification of a supplied hierarchy and efficient generation of one are separate requirements. The present package advances the terminal test and the algebraic route, and gives exact interfaces against which a hierarchy implementation can be audited. It does not supply uniform geometric bounds for  $B, D, C$ . Proving those bounds, together with an executable producer satisfying their hypotheses, remains the step needed for a general hierarchy-based complexity theorem.

## 8 Measurements, validation, and repository integration

The experiments separate three computational questions. The window benchmark measures the cost of obtaining a selected obstruction instead of all homology. The perturbation benchmark computes the same complete homology by two reduction engines. The pattern benchmark compares two classifiers for an already known ball. A speed ratio from one question is not an end-to-end speed ratio for the other two.

All stored measurements use Python 3.12.14 on an x86-64 Linux environment with glibc 2.39. Raw repetitions, input diagrams or deterministic fixture generators, crossing orders where applicable, and operation counts are included. Wall times are observations on this environment. The paired protocol reduces some variability but does not remove shared-machine scheduling noise, especially for millisecond inputs. Asymptotic claims in this paper follow from the preceding proofs, rather than fits to these small timing samples.

### 8.1 Exact windows: less construction, sometimes an earlier verdict

The window experiment uses one warmup per variant and seven timed pairs, alternating which variant runs first. Both variants receive identical PD data and the same preselected crossing order. Both use the baseline’s automatic shape-cache policy. Diagram loading, order selection, and cheap invariant filters are excluded. The window orientation is fixed for each row: the bound-aware automatic mirror selector is available to callers but is not being credited for these measurements.

| Input                | $n$ | Full (ms) | Window (ms) | Full/window | $\dim H^0$ |
|----------------------|-----|-----------|-------------|-------------|------------|
| Conway               | 11  | 7.202     | 7.089       | 1.016       | 10         |
| Kinoshita–Terasaka   | 11  | 8.954     | 7.747       | 1.156       | 10         |
| Hard unknot          | 8   | 1.626     | 1.364       | 1.193       | 2          |
| Determinant-one grid | 10  | 4.826     | 1.455       | 3.316       | 2          |
| Mixed 5-braid        | 36  | 904.254   | 581.562     | 1.555       | 698        |
| Conway, mirrored     | 11  | 13.823    | 13.456      | 1.027       | 10         |
| K–T, mirrored        | 11  | 13.184    | 14.254      | 0.925       | 10         |
| $T(4, 7)$ control    | 21  | 77.047    | 9.280       | 8.303       | 2          |
| $T(2, 31)$ control   | 31  | 11.580    | 6.467       | 1.791       | 2          |

Table 2: Scanner-kernel medians from seven alternating pairs after a warmup of each variant. The window computes only normalized degree zero, while the full scan computes every degree. Input preparation and order selection are excluded.

The most useful case is the repository’s 36-crossing mixed five-braid stress diagram. Its complete unreduced rank is 5898, while its normalized degree-zero rank is 698. Thus this one degree already proves nontriviality. The median full scan took 0.904254 seconds and the window took 0.581562 seconds, a ratio of approximately 1.55. The peak allocated objects decreased from 17694 to 7692, a reduction of about 56.5%. Figure 1 plots every allocation stage, including the large final expansion.

The identical-cap experiment is more informative than a small timing difference. With a 10000-object ceiling the full scan raises `ScanLimit` before allocating 17694 objects, while the window completes with peak 7692 and returns the 698-dimensional obstruction. This is a reproducible change in which exact question can be answered within a stated resource limit. The cap is an object cap, not a measured resident-memory limit; morphisms, dictionaries and caches also use space.

Conway and Kinoshita–Terasaka both have degree-zero dimension 10 in these conventions, so

| Input                | Full peak | Window peak | Window rejects unknot? |
|----------------------|-----------|-------------|------------------------|
| Conway               | 246       | 124         | Yes                    |
| Kinoshita–Terasaka   | 246       | 118         | Yes                    |
| Hard unknot          | 11        | 11          | No                     |
| Determinant-one grid | 54        | 24          | No                     |
| Mixed 5-braid        | 17,694    | 7,692       | Yes                    |
| Conway, mirrored     | 246       | 128         | Yes                    |
| K–T, mirrored        | 246       | 122         | Yes                    |
| $T(4, 7)$ control    | 210       | 40          | No                     |
| $T(2, 31)$ control   | 186       | 66          | No                     |

Table 3: Peak physical objects allocated before elimination. These are object counts, not peak resident-memory measurements. A degree-zero dimension of two is inconclusive.

their window results are also decisive. Their short wall times should not be overinterpreted: the mirrored Kinoshita–Terasaka row is slower with the window. Truncation incurs bookkeeping, and different remaining objects can change the pivot order.

The hard unknot, determinant-one grid knot, and torus controls all have degree-zero dimension 2. Those window results are inconclusive. In particular, the large ratio on the  $T(4, 7)$  control does not constitute a faster complete recognition result: additional homology or a cheap invariant still supplies nontriviality. The standard recognition pipeline already dispatches many of these inputs before Khovanov computation. These rows deliberately isolate the scanner operation.

The saved JSON also contains the isolated-guard pruning ablation. Every ablation agrees on the requested ranks. Its allocation profile explains what this additional operation deletes, but the ablation was not timed and no speed ratio is assigned to it. For the final stress stage the retained allocation is 7692; unit elimination and the valid deletion of isolated guards leave the requested 698 objects.

## 8.2 Perturbation: an exact prototype with a measured tradeoff

The second experiment uses five warmed alternating pairs. The complete scanner and perturbation backend receive the same fixed crossing order, with shape caching disabled in both. They agree on the full homological rank distribution for every row. Optional construction and verification of all contraction maps is excluded from the timing and tested separately.

| Input              | $n$ | Scalar (ms) | Perturbation (ms) | Scalar/pert. |
|--------------------|-----|-------------|-------------------|--------------|
| Hard unknot        | 8   | 2.299       | 3.479             | 0.661        |
| Conway             | 11  | 7.858       | 15.080            | 0.521        |
| Kinoshita–Terasaka | 11  | 8.572       | 19.468            | 0.440        |
| $T(3, 7)$          | 14  | 13.637      | 15.951            | 0.855        |
| $T(4, 7)$          | 21  | 45.236      | 60.884            | 0.743        |

Table 4: Full Khovanov scans: five warmed alternating pairs, fixed order, shape cache disabled in both backends. The perturbation backend is slower on every natural input tested.

The current perturbation implementation loses on every natural diagram in Table 4. The extra binary basis construction and map assembly outweigh the avoided individual cobordism operations in these cases. This is a reason to retain it as an explicit research backend, and to use its rank

profiles and certificates to design a selective hybrid, rather than replace the default elimination routine.

There is nevertheless a regime where the construction has a concrete advantage. Consider two-term algebraic complexes with differential

$$d = I + xR \quad \text{over } \mathbb{F}_2[x]/(x^2),$$

where  $R$  is a deterministic pseudorandom binary matrix. The scalar differential is the identity, so its homology vanishes. Moreover  $(I + xR)^2 = I$ , making full contractibility immediate. These are valid complexes over the one-arc endomorphism algebra, although no claim that they arise from knot diagrams is made.

| $R$ order | Objects | Scalar (ms) | Perturbation (ms) | Scalar/pert. |
|-----------|---------|-------------|-------------------|--------------|
| 16        | 32      | 0.162       | 0.192             | 0.842        |
| 32        | 64      | 0.744       | 0.444             | 1.676        |
| 64        | 128     | 4.022       | 1.449             | 2.776        |
| 128       | 256     | 29.020      | 5.822             | 4.984        |
| 256       | 512     | 204.609     | 31.587            | 6.478        |

Table 5: Synthetic two-term complexes with differential  $I + xR$  over  $\mathbb{F}_2[x]/(x^2)$ . These are algebraic stress tests, not knot diagrams. Both timings include construction of the same input complex.

At matrix order 256, the median decreases from 0.204609 to 0.031587 seconds, about 6.48 times. Both timings construct the same input. The scalar path performs many nilpotent updates; the perturbation path detects scalar acyclicity and returns an empty minimal object. Requiring an explicit contracting homotopy would add work. The benefit therefore supports a specific implementation hypothesis—large scalar rank can make block reduction attractive—without supplying an end-to-end unknot speedup. The stored small stress cases do check the complete contracting identities.

### 8.3 Terminal patterns: a large improvement in a narrow subroutine

| Darts | New (ms) | Report 06 (ms) | Old/new |
|-------|----------|----------------|---------|
| 24    | 0.062    | 2.526          | 40.6    |
| 48    | 0.081    | 25.605         | 315.9   |
| 72    | 0.108    | 152.797        | 1416.0  |
| 96    | 0.149    | 487.551        | 3265.9  |
| 144   | 0.166    | 2745.396       | 16517.9 |
| 192   | 0.440    | 9530.512       | 21650.6 |

Table 6: Known-ball terminal classifier: medians of three complete calls including validation. The fixture family is dual to bipyramid triangulations. No complete hierarchy is timed.

The old and new pattern timings both begin with the same prepared JSON-like input and include validation, spherical-embedding checks, classification, and result construction. Fixture generation and external JSON reading and writing are excluded. The baseline source is the unmodified report-06 implementation, preserved with Git blob hashes. Each median uses three calls.

At 192 darts, the new classifier takes about 0.440 milliseconds and the old bond enumerator about 9.53 seconds. The very large ratio is consistent with eliminating an enumeration of edge subsets

| Darts | New (ms) | Candidate pairs |
|-------|----------|-----------------|
| 384   | 0.526    | 193             |
| 1536  | 5.704    | 769             |
| 6144  | 23.201   | 3073            |
| 24576 | 133.726  | 12289           |
| 98304 | 496.136  | 49153           |

Table 7: Larger essential terminal fixtures, without running the quartic baseline. Counts and individual timings are in the raw JSON.

followed by repeated graph traversal. On the larger fixture with 98304 darts the new classifier takes approximately 0.50 seconds. These fixtures provide useful checks of the linear-space and near-linear-time construction. They give no estimate for normal-surface selection, hierarchy length, or the number of terminal components in a complete recognition algorithm.

## 8.4 Validation and the remaining trust boundary

The delivered fast-scanner suite passes 64 test methods, comprising the 41 baseline methods and 23 additions. The additions cover exact windows, mirror selection and normalization, guarded boundaries, truncation before resource accounting, randomized scan orders, cache choices, full-fallback semantics, the CLI, and finite perturbation identities. The hierarchy suite adds 10 passing methods. The full logs are included; individual methods may contain many exhaustive or randomized cases.

Two especially relevant checks are more discriminating than agreement on one final integer. First, intermediate perturbation models have the same matching-by-degree multiplicities as exhaustive unit cancellation, and their explicit maps satisfy all deformation-retraction identities. Second, all 10410 small rotation maps are censused by the terminal tests, and connected negative witnesses are verified by the old primal-cut checker as well as the new verifier. The tests also reject deliberately corrupted certificates and nonspherical inputs.

The saved HPL example is the unreduced four-crossing prefix of the eight-crossing  $(3, [1, 2]^4)$  braid closure. It reduces 20 source objects to 6 minimal objects at a six-endpoint frontier, and its constructor uses perturbation depth two. Its JSON records the source diagram, actual endpoint pairings, and all five differential and contraction maps. The verifier reconstructs the source prefix and checks the saved maps without recomputing the perturbation construction. A second JSON records a terminal triangle with primal sides of sizes three and three. Both artifacts can be reloaded and checked with `experiments/export_certificates.py--verify-existing`.

The proofs in this article are mathematical arguments accompanied by executable checks. They are not Lean formalizations. The window API returns ranks, a degree convention, and allocation evidence; it does not yet export a short, standalone proof of those ranks checkable without the scanner. The perturbation certificate checks trust the exact implemented cobordism composition. The terminal negative verifier is independent of the triangle-enumeration algorithm but still trusts its input’s known-ball precondition. These are specific interfaces for further formalization, not substitutes for the hypotheses in the correctness theorems.

## 8.5 Integration contract

The complete modified `fast/` directory is included, together with `integration/fast-window.patch`. The patch is based on the active tree at the pinned revision, and its applicability is checked against that exact snapshot. The `report-07` structural patch is a separate lineage; combining both continuations requires an ordinary code integration and the corresponding combined regression run. No remote repository is changed by preparing this package.

The new recognition stage is opt-in. Existing calls preserve their default behavior. The main API additions are:

```
recognize(diagram, use_window=True, window_radius=0,
          window_mirror=True, window_max_objects=10000,
          window_seconds=None)
```

The target is the normalized interval  $[-\text{window\_radius}, \text{window\_radius}]$ . Automatic mirroring compares the exact proved bounds  $B_0, B_1$  and restores the original raw indices before normalization. A result different from  $\{0 : 2\}$  is a nontriviality obstruction. Agreement invokes the full fallback. Exhausting a window-specific budget records a skipped window and leaves the remaining global budget to that fallback. Exhausting the complete budget yields UNKNOWN.

The lower-level routines distinguish raw-degree queries from this normalized decision interface:

```
from fastunknot.window_scan import khovanov_window
from fastunknot.window_bounds import khovanov_window_auto

# Raw degrees, fixed orientation:
part = khovanov_window(diagram.pd, lower=a, upper=b,
                       order=order, max_objects=10000)

# The same original raw degrees, selecting a mirror internally:
part = khovanov_window_auto(diagram, a, b, order=order)
```

The result field is `window_rank`. A separate `complete_rank` flag is true only when the requested interval covers the entire possible raw range  $[0, n]$ . The `window` CLI subcommand also accepts raw bounds, recorded as `raw_lower` and `raw_upper`; it does not label a partial answer as a total rank. The complete recognizer's `-window` flag enables the optional obstruction stage.

The module `perturbation.py` is available for direct research use and certified intermediate reductions; it is not selected by the default recognizer. The `hierarchy/` directory is a self-contained terminal classifier and experiment bundle. It is intended for a future hierarchy implementation or a report appendix with its ambient-ball contract intact.

**Reproducibility artifacts.** The package includes the TeX source and figure PDFs, raw timing JSON, deterministic fixture generators, source provenance, the integration patch, and complete tests. The `experiments/` scripts regenerate benchmark data, figures, tables and example certificates. Figure generation checks the full allocation trace against the saved stress rank and peak before plotting it. The tables are generated directly from the raw JSON so that the numerical claims can be updated without manual transcription.

These deliverables support two immediate engineering choices: enable the window obstruction for workloads where it can avoid an expensive full scan, and replace repeated terminal bond enumeration

where the known-ball precondition is already certified. The nilpotent backend supplies a separate algebraic foundation for the next optimization cycle.

## 9 Prioritized research questions

The next investigations should target a quantified obstruction or a missing executable operation. A useful outcome may be a positive theorem, a counterexample that rules out a proposed parameterization, or an exact implementation with a proved narrower domain. The following questions are ordered by their proximity to the present algorithms and their potential to close the remaining complexity gap.

**1. How far from degree zero must a knot witness occur?** For a nontrivial knot  $K$ , define its homological witness radius by

$$d_{\text{wit}}(K) = \min \left\{ d \in \mathbb{Z}_{\geq 0} : \exists h \in [-d, d], \dim_{\mathbb{F}_2} H_{\text{norm}}^h(K) \neq 2 \mathbf{1}_{\{h=0\}} \right\}.$$

The full-rank detection criterion makes this finite, but the present paper gives no useful uniform bound. Seek an explicitly recognizable family of diagrams for which  $d_{\text{wit}}(K) = O(\log n)$  and for which the geometric conditions in Corollary 3.7 can be achieved algorithmically. Such a theorem would turn agreement with one specified window into a complete decision on that family. A counterfamily with a large witness radius would identify the necessary extra parameter. Tests should include knots whose degree-zero homology agrees with the unknot; a successful degree-zero rejection set alone cannot answer the question.

**2. Can geometric assembly remove the all-state circle factor?** The bound in Theorem 3.5 deliberately allows arbitrary valid scan frontiers and pays for the number of circles in an extreme smoothing. Kelomäki–Schütz obtain stronger extremal object bounds under their geometric scanning hypotheses [3]. The concrete problem is to construct the required assembly from the current PD input, retain the moving interval and its differential guards, and prove a bound for every intermediate allocation in that assembly. It is also necessary to track the relation between its raw degrees, normalized degrees, and mirror choice. The desired result is an executable construction with a quantified frontier and circle bound, rather than an assumption that an arbitrary crossing order is geometrically nice.

**3. Which intrinsic multiplicity profiles admit compact continuation?** Theorem 4.3 makes the numbers  $m_{a,h}$  independent of the chosen minimal model at a fixed frontier. Consequently a different scalar pivot order cannot reduce those intrinsic multiplicities. The relevant question is whether their repeated structure, together with every differential incidence, admits a smaller description preserved by crossing extension and reduction. Candidates include matrix recurrences, shared subcomplexes with shift multiplicities, and arithmetic circuits for repeated coefficient blocks. A theorem should bound the encoded description, the cost of each continuation, and the cost of the final decision query. Bounding only the number of matching labels leaves the known multiplicity obstruction untouched. A first target is a prime-knot family whose minimal object count grows rapidly but whose continuation has a provably small recurrence.

**4. When can the finite perturbation lift be performed by blocks?** The cost in (19) allows dense maps. Natural scan complexes may contain smaller scalar blocks whose interaction through positive-degree morphisms has additional structure. Can one partition these blocks so that the cost of all correction terms is bounded by a sum of local block costs and a small interface cost? The proof must retain the induced maps between blocks: deleting an apparently isolated scalar block can create a nonzero correction elsewhere. A useful parameter would bound the support growth of successive products  $(h\delta)^j$  for  $j \leq w$ . Experiments should compare the complete allocation, reduction, and certificate cost on natural knot complexes, and distinguish those results from deliberately favorable synthetic blocks.

**5. Can one certified categorical interface cover every scanner mode?** The moving-window and perturbation proofs both depend on exact functorial operations and explicit chain homotopies. Develop a common finite interface containing matching objects, oriented cobordism basis elements, composition, crossing-extension functors, and contraction certificates. A Lean development can first formalize the positive grading, its nilpotence consequence, and the bounded-continuation support lemma, then connect these theorems to a checker for finite matrix data. The endpoint-orientation condition in Lemma 4.1 must appear as a hypothesis; unrestricted abstract perfect matchings are not an interchangeable input category. The objective is a small semantic checker shared by full, windowed, block, and structurally compressed computations.

**6. What is the exact compressed-cut output type?** Implement a first complete special case of compressed normal-surface cutting with an output contract that includes the nonproduct core, product-region multiplicities, bundle sidedness, attachment maps, and boundary-pattern data. The family in Proposition 6.1 is a mandatory binary-weight test: increasing  $W$  must not create  $W$  output records. Certified parallel copies of a connected two-sided surface give one initial domain on which the compression is exact. Extension beyond that domain must handle one-sided base surfaces and disconnected primitive coordinate vectors. State the complexity in the input handle count, coordinate bit lengths, and compressed output size, and specify which queries on an aggregated family remain efficient.

**7. Can a terminal obstruction be transported without expansion?** The new terminal verifier supplies a short dual cycle and its primal edge-cut interpretation. Construct the next missing operation: carry that curve backwards across one certified cut or bundle simplification, producing either a violating disc for the preceding pattern or a compression or pattern-compression disc for an earlier surface. The record must identify the surface being compressed, retain the boundary arcs and their orientations, and preserve the relevant intersection count. Prove a bound on encoded size under a sequence of such transfers. This would connect the implemented terminal algorithm to actual hierarchy revision. Rechecking a graph witness at the end does not establish this transport operation.

**8. Can actual restart choices satisfy a small branching budget?** Apply Theorem 7.1 to a fully specified producer, rather than to a hypothetical collection of good hierarchies. The main target is a geometric invariant proving  $\sum \log b(v) = O(\log^2 n)$  on every explored path, with polynomial depth and per-node cost. A weaker  $O(\log^3 n)$  theorem would still give a general quasi-polynomial bound. Forced bundle reductions should contribute to depth and work without being counted as artificial branching. Conversely, repeated failed choices, rebuilding a discarded suffix, and changing

global Heegaard data must all be represented. An informative negative result would exhibit a family where a proposed budget stays small on successful certificates but grows on unavoidable failed continuations.

**9. Can surface selection enforce the needed progress bounds?** A nonzero cohomology class already supplies a route to a nonseparating surface. The next question is to select a surface with quantitatively favorable pattern complexity and subsequent progress, within a proved bit-cost bound. Polynomially checkable optimal-surface certificates do not themselves supply that selection procedure. Investigate restricted handle structures on which controlled homology classes, simultaneous surfaces, or a provably small candidate family achieves the required quality. Every selection rule must account for pattern intersections, not just Euler characteristic or genus. Its guarantees must survive the compressed cuts and witness-driven replacements performed by the same engine. A useful first theorem would identify an invariant class of states on which cheap selection also implies a strictly quantified decrease in the proposed global resource.

**10. Can an exact portfolio inherit the best proved backend bound?** The present recognizer combines necessary obstructions, window queries, and a complete fallback. A future hierarchy engine and structured minimal-complex engine will add further choices. Specify a deterministic budget schedule and shared certificate format such that, on any input where one backend has a proved time bound, the portfolio finishes within a controlled factor of that bound plus explicit preprocessing cost. Already computed degree windows should be reused only when their guard and continuation hypotheses remain valid. Memory ceilings, certificate verification, and repeated state construction belong in the accounting. This problem connects practical dispatch to a precise complexity statement without declaring a window agreement or an exhausted budget to be a mathematical verdict.

### 9.1 Milestones that would change the global claim

There are two concrete ways to strengthen the package’s main recognition guarantee. On the algebraic side, a witness-radius theorem combined with an executable parameter-controlled diagram presentation would make the prescribed quasi-polynomial window complete on its stated input class. A representation theorem for intrinsically large minimal complexes could enlarge that class further.

On the geometric side, the decisive milestone is a producer with compressed cutting, certified witness transport, and proved bounds for the actual execution quantities  $B, D, C$  in Corollary 7.2. The terminal improvement already has its own correctness and complexity theorem, but it cannot replace these global estimates. Either route should culminate in a theorem whose hypotheses are checked by the delivered code or stated as an explicit input promise, accompanied by a complete certificate on every reported decision.

## References

- [1] Vladimir Reshetnikov and ProveIt contributors. *ProveIt: Topology/UnknotRecognition*. Active scanner, reports 06 and 07, and associated research artifacts. Audited revision ca61a1a5f967999d778d6efd7c882e77a2eb379d, 7 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/ca61a1a5f967999d778d6efd7c882e77a2eb379d/Topology/UnknotRecognition>.
- [2] Dror Bar-Natan. *Fast Khovanov homology computations*. Preprint, 2006; published in Journal of Knot Theory and Its Ramifications, 16 (2007), 243–255. <https://arxiv.org/abs/math/0606318>. Author’s accompanying material: <https://www.math.utoronto.ca/~drorbn/papers/FastKh/>.
- [3] Tuomas Kelomäki and Dirk Schütz. *On computational complexity of Khovanov homology*. Preprint, 5 January 2026, arXiv:2601.02119v1. In particular Theorems 1.2–1.4, Proposition 1.5, and Section 5. <https://arxiv.org/abs/2601.02119v1>.
- [4] Marc Lackenby. *Incompressible surfaces, hierarchies and unknot recognition*. Preprint, 25 July 2026, arXiv:2607.23350v1. In particular Proposition 2.6, Proposition 9.1, Proposition 10.2, Proposition 14.2, and Theorem 14.4. <https://arxiv.org/abs/2607.23350v1>.
- [5] Marc Lackenby. *Unknot recognition in quasi-polynomial time*. Public slides, February 2021, hosted by the University of Oxford. Slides 13–14 (PDF pages counted from one) display  $n^{O(\log n)}$ . <https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk.pdf>.
- [6] Marc Lackenby. *Unknot recognition is quasi-polynomial time*. Seminar announcement and abstract, Yau Mathematical Sciences Center, Tsinghua University, 7 November 2024 (announcement dated 6 November). The abstract displays  $2^{O((\log n)^3)}$ . <https://qzc.tsinghua.edu.cn/en/info/1122/4171.htm>.
- [7] OpenAI. *Parabolic Intersections in Artin Groups*. Repository manuscript, 23 September 2026, openai/math; audited revision adc7f1241b42e322a6451854ab7e4b4c146bf78a. The source reviewed here is the lemma “Composition at the bottom” and its bounded-layer construction, together with the categorical-action section; no other claim of the manuscript is required. [https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Parabolic-intersections-in-Artin-groups-September-23-2026/build/sections/03\\_layers.tex](https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Parabolic-intersections-in-Artin-groups-September-23-2026/build/sections/03_layers.tex). Main source: build/paper.tex in the same manuscript directory.
- [8] P. B. Kronheimer and T. S. Mrowka. *Khovanov homology is an unknot-detector*. Publications Mathématiques de l’IHÉS 113 (2011), 97–208; preprint arXiv:1005.4346, 2010. <https://arxiv.org/abs/1005.4346>.
- [9] Alexander N. Shumakovitch. *Torsion of the Khovanov homology*. Preprint arXiv:math/0405474, 2004; version 2, 19 June 2018. Corollary 3.2.C supplies the reduced/unreduced splitting over  $\mathbb{F}_2$ . <https://arxiv.org/abs/math/0405474v2>.
- [10] Marius Crainic. *On the perturbation lemma, and deformations*. Preprint, 2004, arXiv:math/0403266. <https://arxiv.org/abs/math/0403266>.
- [11] Mikhail Khovanov. *A functor-valued invariant of tangles*. Preprint, 2001, arXiv:math/0103190. <https://arxiv.org/abs/math/0103190>.